

Literature Review

Mammographic Breast Cancer Detection with Conformal Uncertainty Quantification

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Abstract

This review surveys the research landscape underpinning deep learning system for breast cancer detection from digital mammography, organised around the two methodological commitments that define the proposed work: the fusion of bilateral information through an explicit asymmetry signal, and the quantification of predictive uncertainty through distribution-free conformal prediction. Rather than cataloguing prior systems individually, The review develops the conceptual landscape from which these commitments emerge, the architectural and clinical foundations of mammographic deep learning, the progression from single-view through dual-view to bilateral modelling, and the shift from heuristic confidence estimates toward formal coverage guarantees and locates, within that landscape, the gap that the proposed system is designed to fill.

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1 Introduction

Breast cancer remains the most commonly diagnosed malignancy among women worldwide, and mammographic screening is the primary tool through which it is detected early enough for intervention to alter outcomes. The clinical value of screening, however, is bounded by the limits of human reading: sensitivity varies between radiologists, declines with fatigue across high-volume screening sessions, and falls sharply when lesions are obscured by radiographically dense tissue. It is against these limits that deep learning has been positioned, with the most prominent systems now reporting diagnostic performance comparable to, and in controlled studies exceeding, that of experienced radiologists [1], [2].

Progress at the level of headline performance has nonetheless left two capabilities underdeveloped, and it is these two that the present work takes as its subject. The first concerns how information from a screening examination is combined. A standard mammographic study images each breast from two angles, yet the diagnostic logic that radiologists apply to those images — the habitual comparison of left against right, in which a region that departs from its contralateral counterpart draws immediate suspicion — is only partially reflected in systems that process views or breasts in isolation. The second concerns how confidence is expressed. A screening classifier that is wrong while appearing certain is more dangerous than one that signals its own doubt, yet the uncertainty estimates accompanying most deployed systems rest on heuristic foundations that carry no formal guarantee.

This review establishes the conceptual context for both concerns. It traces the architectural and clinical foundations on which mammographic deep learning rests, follows the progression from single-view through dual-view toward genuinely bilateral modelling, and examines the move from heuristic confidence toward the distribution-free guarantees offered by conformal prediction. The intent throughout is not to enumerate prior systems but to develop the landscape they collectively define, and so to make visible the specific gap — the union of bilateral asymmetry fusion with conformal uncertainty quantification — that the proposed system occupies. Citations follow the numeric convention; references are listed in full at the close.

2 Foundations of Mammographic Deep Learning

The deep learning systems that now dominate mammographic analysis inherit their representational machinery from a small lineage of convolutional architectures, and the particular member of that lineage adopted here is the densely connected network, in which each layer draws its input from the concatenated outputs of all layers preceding it [3]. The appeal of this connectivity pattern in a medical imaging setting is practical rather than incidental: it sustains gradient flow through deep stacks, encourages the reuse of features across layers, and reaches competitive accuracy at a parameter budget modest enough to be fine-tuned on datasets far smaller than those for which such networks were first devised. That last property is decisive in mammography, where annotated data is scarce relative to the natural-image corpora on which backbones are pre-trained, and where the regularising effect of a compact, reuse-heavy architecture translates directly into more reliable transfer.

The credibility of applying such architectures to screening at all was established less by any single methodological advance than by demonstrations of performance at clinical scale. The most visible of these evaluated a system across screening populations drawn from more than one healthcare system and reported reductions in both false positives and false negatives relative to the standard of care, alongside a reading study in which the system outperformed every radiologist enrolled [1]. Read narrowly, such results are a performance claim; read more broadly, they fixed the expectations against which subsequent mammographic systems are now measured, and they are the reference point any new architecture is implicitly asked to justify itself against. A parallel line of academic work, conducted at comparable scale but with its methodology and code placed in the open, supplied what the proprietary demonstrations could not [2]: a reproducible account of how a mammographic classifier processes the multiple views of a screening study and combines them into a case-level judgement. It is this account, rather than the headline performance figures, that supplies the methodological baseline for the present work.

Underlying both lines of work is a dataset dependency that shapes what is practically buildable. The curated screening subset most widely used for open mammographic research provides expert-annotated cases with pathology labels, BI-RADS assessments, and lesion-level

delineations [4], and its structure is consequential in ways that are easy to underestimate. Cases are organised such that the two breasts of a patient, and the two standard views of each breast, must be reassembled by joining records on patient identity and laterality before any bilateral relationship can be modelled at all; and because the canonical train–test partition is defined at the level of the case rather than the image, careless handling readily leaks one side of a patient into training while the other is held out for test. The bilateral construction on which the proposed system depends is therefore conditioned, from the outset, on the organisation of this dataset, which makes its structure a matter of architectural rather than merely administrative concern.

3 From Single Views to Bilateral Asymmetry

The trajectory of mammographic modelling can be read as a steady widening of the field of view that the classifier is permitted to consider at once. The reproducible academic baseline already noted treats the four images of a screening study together, weighting their contributions through learned attention rather than collapsing them prematurely [2], and the immediate successor to that design made the relationship between the two views of a single breast explicit, processing the craniocaudal and mediolateral oblique projections jointly so that evidence visible in one projection could inform interpretation of the other [5]. That dual-view design is the closest architectural antecedent to the present work, and the distinction between them is precisely the axis along which the proposed system advances: where the antecedent fuses the two views of the same breast, the proposed system additionally fuses across breasts, treating the contralateral side not as a separate case to be classified but as the reference against which asymmetry is measured.

The diagnostic motivation for that additional axis is well established in the modelling literature, where the difference between corresponding left and right representations has been used directly as a discriminative signal. The most direct precedent computes an explicit asymmetry feature as the absolute difference between the encoded representations of the two breasts produced by a shared-weight encoder, and shows that supplying this signal to the classifier improves discrimination over an otherwise identical unilateral baseline [6]. The proposed system

takes this construction as its point of departure, retaining both the shared-weight encoder and the difference signal $\delta = |f_L - f_R|$, and extends it with a gating mechanism that allows the magnitude of the asymmetry to modulate how strongly each view representation contributes to the final judgement.

That asymmetry can be exploited in more than one way is itself instructive. A contrasting approach uses the bilateral relationship generatively, synthesising a plausible healthy counterpart for a suspicious breast and classifying by comparison against that counterfactual [7]. The shared premise — that the contralateral breast carries diagnostic information no single image contains — is exactly the premise the proposed system adopts; the divergence lies in mechanism, and the divergence matters. A generative counterpart introduces an auxiliary synthesis objective and the attendant training complexity, and renders the basis of a decision harder to inspect. By keeping asymmetry as a direct, differenced feature fused through attention, the proposed system retains a representation whose contribution can be visualised by standard gradient-based attribution, a property of considerable value where a screening decision may ultimately require justification to a clinician.

4 From Confidence to Coverage

Quantifying how far a prediction may be trusted is, in a screening context, not an adjunct to the classification problem but part of it. The estimates of uncertainty that accompany most deployed systems are heuristic in origin — derived from the softmax response, from stochastic forward passes, or from ensemble disagreement — and while such estimates are often empirically useful, they share a defining limitation: none carries a guarantee about the rate at which the truth will actually fall within the range the model declares. Conformal prediction occupies a different footing. In its split formulation, a held-out calibration set is used to convert a model’s raw scores into prediction sets that satisfy a finite-sample coverage guarantee, ensuring that for any chosen error level the true label is contained with at least the corresponding probability, and doing so without assumptions on the underlying data distribution [8]. For a screening system, the consequence is concrete: rather than emitting a single label and an uncalibrated confidence, the classifier can return a set whose validity is guaranteed in a precise and verifiable sense,

contracting to a single class where the evidence is clear and expanding to admit both possibilities where it is not.

The naive form of this construction, however, can yield sets larger than necessary, diluting the practical value of the guarantee. A refinement addresses this directly by adding a regularisation term that penalises the inclusion of implausible classes, producing sets that remain valid in the coverage sense while being substantially more economical in size [9]. It is this refined construction that the proposed system adopts, calibrating it on a dedicated split so that the prediction sets it returns are both formally guaranteed and clinically legible: a singleton when the model can commit, and a two-element set — whose natural reading is a recommendation to recall for further assessment — when it cannot.

5 Synthesis

Surveyed together, these strands describe a landscape with a definite shape and a definite vacancy. The architectural foundations supply a backbone suited to data-scarce transfer [3]; the clinical and reproducible baselines establish both the performance expectations of the field and a concrete account of multi-view processing [1], [2]; the dataset on which open research depends conditions what bilateral modelling requires in practice [4]. Along the modelling axis, the progression runs from multi-view [2], through dual-view [5], to the explicit use of bilateral asymmetry as a differenced feature [6] or as a generative prior [7]. Along the uncertainty axis, the movement is from heuristic confidence toward distribution-free guarantees [8], made practical by size-regularised prediction sets [9].

What no single line within this landscape provides is their union. The bilateral systems quantify no uncertainty in any guaranteed sense; the conformal methods are developed without reference to the bilateral structure of mammography; and the closest architectural antecedent stops at the boundary between the two views of one breast, short of the contralateral comparison that defines clinical reading. The proposed system is situated precisely at that intersection, joining an explicit, attention-gated bilateral asymmetry signal to a distribution-free conformal guarantee within a single architecture trained on openly available data — a combination that the surveyed literature motivates at every point but nowhere assembles.

6 Conclusion

The foregoing has developed, rather than enumerated, the context for the proposed system. From the architectural lineage it draws a backbone matched to the data regime of mammography; from the clinical and academic baselines it inherits both a standard of performance and a reproducible model of multi-view processing; from the asymmetry literature it takes the differenced bilateral signal it extends, and the generative alternative it deliberately declines; and from the conformal literature it adopts the guarantee, and the refinement of that guarantee, on which its treatment of uncertainty rests. The scope of the review has been kept deliberately narrow, admitting only those works that bear directly on the two commitments the system is built around, so that the gap it occupies — the conjunction of bilateral asymmetry fusion with conformal uncertainty quantification — stands clear of the surrounding field.

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